

Analyzing Time-of-Day Demand and Payment Patterns to Optimize Operations at Balaji Fast Food

A Proposal report for the BDM Capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled "Analyzing Time-of-Day Demand and Payment Patterns to Optimize Operations at Balaji Fast Food". As this project is conducted under the BDM Capstone Project - Analysis with Secondary Data pathway, I extend my appreciation to the creators of the publicly available "Balaji Fast Food Sales" dataset on Kaggle, whose data has enabled me to conduct this project.

I hereby assert that this project uses secondary data obtained from a publicly available Kaggle repository, and not primary data collected directly from a business. This has been explicitly disclosed here and in the relevant sections of this report, in accordance with the BDM Capstone Project - Analysis with Secondary Data guidelines.

I affirm that all procedures employed for data cleaning and analysis have been duly explained in this report, and that the outcomes and inferences presented are an accurate depiction of the findings obtained through the analytical procedures described.

I am dedicated to upholding academic honesty and integrity, and I am receptive to any additional examination or validation of the data and analysis contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all work undertaken has been solely conducted by me. In the event that plagiarism is detected in this report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all recommendations are specific to the dataset/business context analyzed in this project and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate: (Digital Signature)

Satyam Sharma

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Date: 21 July 2026

1 Executive Summary and Title

This project, titled "Analyzing Time-of-Day Demand and Payment Patterns to Optimize Operations at Balaji Fast Food", analyzes secondary sales-transaction data of Balaji Fast Food, a quick-service fast-food outlet. The dataset source is the publicly available "Balaji Fast Food Sales" dataset on Kaggle (kaggle.com/datasets/ahmedhalimo/balaji-fast-food-sales), comprising 1,000 order records collected between April 2022 and March 2023, with fields for item purchased, item category, quantity, item price, transaction value, payment mode, time of sale, and the staff member handling the transaction.

The outlet currently lacks a structured understanding of when specific items are in demand and how payment preferences influence revenue collection. Menu-preparation and staff-scheduling decisions appear to be made without reference to time-of-day demand patterns, while continued reliance on cash transactions (58% of orders by value) increases the risk of reconciliation errors and limits digital revenue traceability.

The analysis will be carried out in Python, using pandas for data cleaning and manipulation (handling currency-formatted fields, parsing dates, removing duplicates) and matplotlib/seaborn for visualization. Specific techniques include descriptive statistics, cross-tabulation of time-of-sale against item and item-type, revenue-contribution (ABC-style) ranking of items, and a chi-square test of independence to check whether payment method is associated with time slot or item type. Key aggregates will be cross-verified in Excel.

The expected outcome is a data-backed staffing and food-preparation guideline aligned with time-of-day demand, together with a recommendation to encourage digital payments where cash dependence is highest, helping the outlet operate more efficiently without additional capital investment.

2 Organisation Background

The organization under study, Balaji Fast Food, is a small quick-service restaurant (QSR) operating in the unorganized food and beverage retail segment. It sells ready-to-eat, street-food-style snacks — Panipuri, Vadapav, Aalopuri, Frankie, and Sandwich — along with beverages such as Sugarcane Juice and Cold Coffee. The outlet operates on a B2C model, serving walk-in retail customers across five time slots through the day: Morning, Afternoon, Evening, Night, and Midnight.

As this project uses secondary data obtained from a public Kaggle repository, organizational details such as exact ownership, location, and years of operation are not disclosed in the source dataset. However, the transaction-level data available — 1,000 orders spanning April 2022 to March 2023, with order ID, item name, item type, price, quantity, transaction amount, payment type, staff gender, and time of sale — provides sufficient granularity to study demand and revenue patterns representative of a typical small fast-food business, and forms the basis for this analysis.

3 Problem Statement (Listed as objectives)

3.1 Objective 1

To identify item-wise and time-of-day demand and revenue patterns at Balaji Fast Food, in order to guide staffing and food-preparation planning.

3.2 Objective 2

To evaluate the impact of payment method (Cash vs. Online) on transaction value and order volume, and to recommend strategies to encourage digital payment adoption where cash dependence is highest.

4 Background of the Problem

Small and unorganized quick-service food outlets such as Balaji Fast Food typically operate without structured data-analytics support. Purchase quantities, staffing rosters, and food-preparation schedules are largely decided based on the owner's intuition rather than historical demand patterns. This often results in two opposing inefficiencies: under-preparation during genuinely high-demand time slots, causing lost sales, and over-preparation during low-demand slots, leading to wastage of perishable ingredients and unnecessary staffing costs.

A related issue in such businesses is continued dependence on cash transactions. While cash offers convenience to some customer segments, it also complicates daily reconciliation, increases the risk of under-reporting, and limits the business's ability to build a verifiable digital sales trail that could later support credit access or aggregator tie-ups.

Both issues stem from a common root cause: the absence of a systematic method to convert raw, item-level transaction records into decision-ready insights. Internal causes include limited technical know-how and time constraints typical of small-business owners; external causes

include the scarcity of affordable analytics support tailored to micro and small food businesses. This project addresses this gap using the outlet's own transaction data.

5 Problem Solving Approach

The approach for this project follows four broad phases: data cleaning and preparation, descriptive analysis, objective-aligned pattern analysis, and synthesis of recommendations.

Data cleaning and preparation: The raw dataset (1,000 records) will be checked for missing values, duplicate order IDs, and formatting inconsistencies — notably, currency symbols embedded within the `item_price` and `transaction_amount` fields, which must be stripped and converted to numeric type before analysis. Date fields will be parsed into a proper datetime format to enable month-wise and day-wise aggregation where relevant.

Descriptive statistics: Measures such as mean, median, and distribution of item price, quantity, and transaction amount will be computed to establish the overall sales profile — total revenue, average order value, and order volume split by item category (Fastfood vs. Beverages).

Time-of-day and item-wise demand analysis (Objective 1): A cross-tabulation of `time_of_sale` against `item_name` and `item_type` will be built to identify which items see higher demand during specific time slots. This will be visualized using grouped bar charts and heatmaps. A revenue-contribution analysis will additionally rank items by their share of total revenue, combined with the time-slot dimension rather than a plain single-dimension Pareto split, in line with the guidance to avoid superficial 80/20 conclusions that add little value.

Payment-method analysis (Objective 2): Transactions will be grouped by `transaction_type` (Cash vs. Online) to compare average transaction value, order volume, and revenue share. A chi-square test of independence will be used to check whether payment method is associated with item type or time slot, helping determine whether specific slots or products are disproportionately cash-driven.

Tools: The analysis will be carried out in Python (pandas for data manipulation; matplotlib/seaborn for visualization), with key aggregates cross-verified in Excel. Python is preferred for this dataset size because it allows reproducible, scriptable cleaning and cross-tabulation, and integrates statistical tests such as chi-square directly into the workflow.

The insights from these analyses will be triangulated into specific, actionable recommendations on staff scheduling, prep-quantity planning by time slot, and a phased strategy to increase digital-payment adoption.

6 Expected Timeline

6.1 Work Breakdown Structure

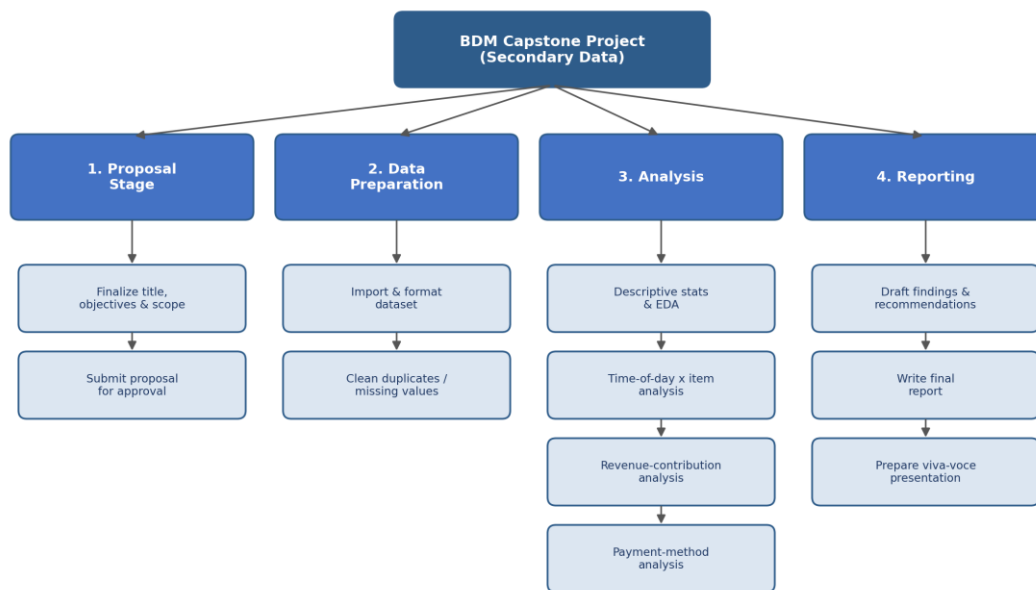


Figure 1: Work Breakdown Structure for the project

6.2 Gantt Chart

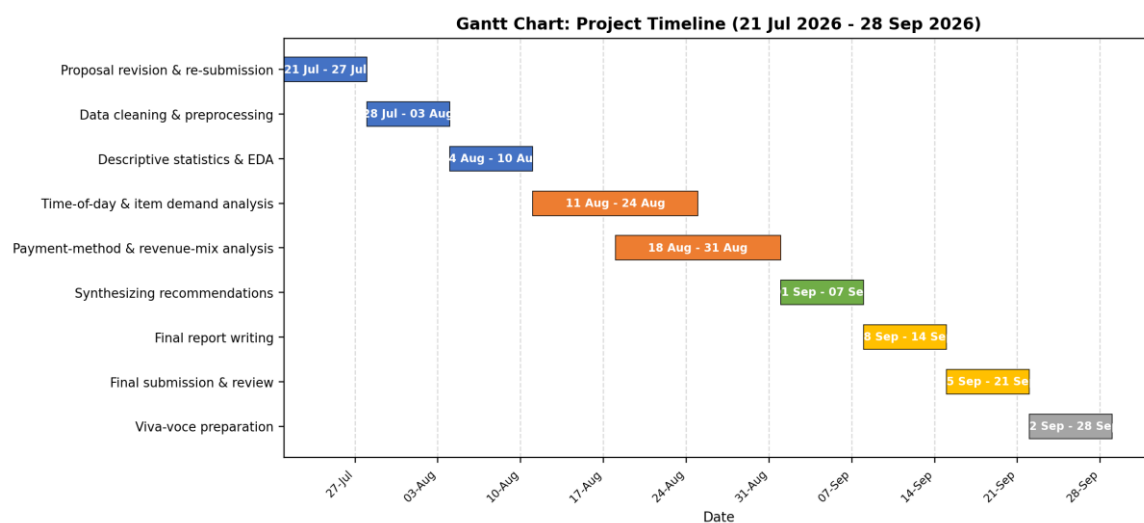


Figure 2: Gantt chart with dated task schedule (21 Jul 2026 - 28 Sep 2026)

Task	Description	Start Date	End Date
Proposal revision & re-submission	Address evaluator feedback (declaration signature, executive summary detail, timeline format) and resubmit proposal for approval.	21-Jul-2026	27-Jul-2026
Data cleaning & preprocessing	Import the 1,000-row dataset, strip currency symbols from price/transaction fields, parse dates, and remove duplicates/missing values.	28-Jul-2026	03-Aug-2026
Descriptive statistics & EDA	Compute mean/median/distribution of price, quantity and transaction value; establish baseline revenue and volume profile by item category.	04-Aug-2026	10-Aug-2026
Time-of-day & item demand analysis	Cross-tabulate time-of-sale against item/item-type; build heatmaps and a revenue-contribution ranking to locate peak demand periods.	11-Aug-2026	24-Aug-2026
Payment-method & revenue-mix analysis	Compare cash vs. online transactions on value, volume and revenue share; run a chi-square test against time slot and item type.	18-Aug-2026	31-Aug-2026
Synthesizing recommendations	Consolidate demand and payment-method findings into specific staffing, prep-planning and digital-payment recommendations.	01-Sep-2026	07-Sep-2026
Final report writing	Compile the complete 18-20 page final report, including results, interpretation and recommendations.	08-Sep-2026	14-Sep-2026
Final submission & review	Submit the final report through the project portal and await evaluator review/approval.	15-Sep-2026	21-Sep-2026
Viva-voce preparation	Prepare a maximum-10-slide presentation summarizing key findings for the viva-voce examination.	22-Sep-2026	28-Sep-2026

7 Expected Outcome

Mapped against the two identified problems (demand-blind staffing/prep planning, and cash-dependent, reconciliation-risky payments), this project is expected to deliver the following outcomes:

- Time-slot-wise demand map: a clear view of which time slots (Morning, Afternoon, Evening, Night, Midnight) see peak order volume, to guide staff-shift planning.
- Item-wise demand map by time slot: identification of which specific items peak in which slots, enabling targeted ingredient prep rather than uniform stocking.
- Reduced stock-out risk: fewer lost sales during genuine high-demand periods, since prep quantities will be aligned with historical demand rather than intuition.
- Reduced wastage: lower over-preparation of perishable items during predictably slow slots, cutting unnecessary ingredient cost.
- Revenue-contribution ranking of the menu: a ranked view of which items/categories (Fastfood vs. Beverages) drive the most revenue, to inform pricing and promotion decisions.
- Payment-mix diagnosis: quantified evidence of whether specific time slots or item types are disproportionately cash-paid, based on the chi-square association test.
- Digital-payment strategy: a targeted (not blanket) recommendation — e.g., digital-payment incentives during identified high-cash-usage slots — to reduce reconciliation risk and improve revenue traceability.

Collectively, these outcomes are expected to help the outlet improve operational efficiency and reduce revenue-reconciliation risk without requiring additional capital investment.